YISHUN INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE NAME	
CG	INDEX NO

PHYSICS 9478/02

Paper 2 Structured Questions **25 August 2026**

You must answer on the question paper. **2 hours**
No Additional Materials are required.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, CG and index number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- You may use an approved calculator.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

For Examiner's Use	
Paper 1	
	/30
Paper 2	
1	**/12**
2	**/7**
3	**/10**
4	**/6**
5	**/9**
6	**/10**
7	**/21**
Penalty	
Total	**/75**

This document has **24** pages. Any blank pages are indicated.

Data

speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}\,\text{H m}^{-1}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$
	$(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1})$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
unified atomic mass constant	$u = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31}\,\text{kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27}\,\text{kg}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Avogadro constant	$N_A = 6.02 \times 10^{23}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{GM}{r}$
temperature	$T/\text{K} = T/^\circ\text{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas particle	$E = \frac{3}{2}kT$

displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm \omega \sqrt{(x_0^2 - x^2)}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots$
capacitors in parallel	$C = C_1 + C_2 + \ldots$
energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_0 e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2}n^2$
radioactive decay	$x = x_0 e^{-\lambda t}$
radioactive decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Answer **all** the questions in the spaces provided.

1 **(a)** One end of a cable is connected to a fixed point. A load is attached to the other end so that the wire hangs vertically.

The initial unstretched length L, the diameter d of the cable, the applied load F and the resulting extension x are measured as

$L = 2.000 \pm 0.002$ m

$d = 0.40 \pm 0.02$ mm

$F = 25.0 \pm 0.5$ N

$x = 1.20 \pm 0.06$ mm

The Young modulus E of the material of the cable is calculated by using the expression

$$E = \frac{4FL}{\pi d^2 x}$$

Calculate E and the uncertainty in its value.

Show how your answers were obtained.

E = …… ± …… N m^{-2} [4]

(b) A uniform rectangular sign of length 0.80 m and width 0.68 m is fixed at its corner points A and B to a wall. The sign is also supported by a wire CD, as shown in Fig. 1.1

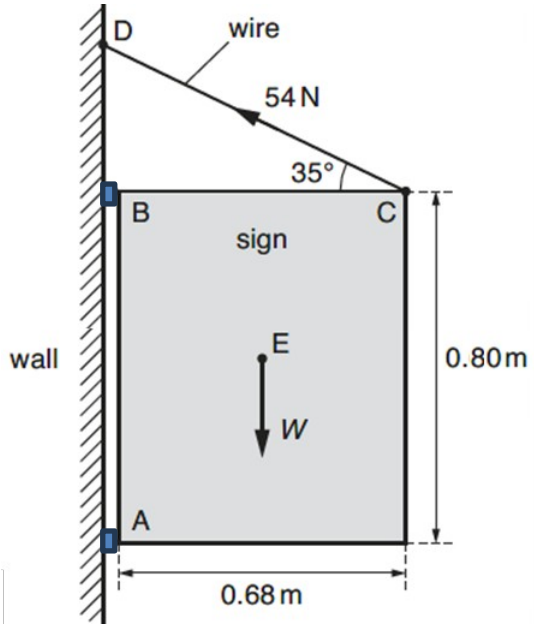


Fig. 1.1

The sign has weight *W* and centre of gravity at point E. The sign is held in a vertical plane with side BC horizontal. The wire is at an angle of 35° to side BC. The tension in the wire is 54 N.

The force exerted on the sign at B is only in the vertical direction.

(i) State two conditions necessary for the sign to be in equilibrium.

……

……

……

…… [2]

(ii) Explain why the force on the sign at B does not have any moment about point A.

……

…… [1]

(iii) Determine the weight *W* of the sign.

W = …… N [2]

(iv) Calculate the total vertical force exerted by the wall on the sign at points A and B.

total vertical force = …….. N [2]

(v) The attachment at corner point B is removed but the sign remains stationary, supported solely by the wire CD and its remaining contact with the wall at corner point A.

On Fig. 1.1, draw an arrow to show the direction of the force exerted by the wall on the sign at point A. [1]

[Total: 12]

2 Two positively charged particles X and Y of masses 3*m* and 2*m* respectively, move directly towards each other with the same speed as illustrated in Fig. 2.1.

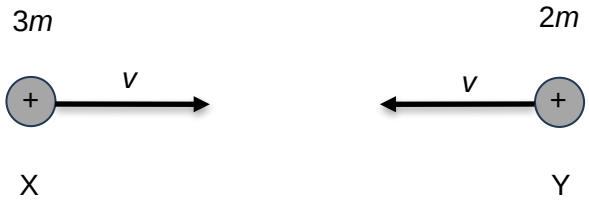


Fig. 2.1

Fig. 2.2 shows how the velocity of each particle varies with time during the collision.

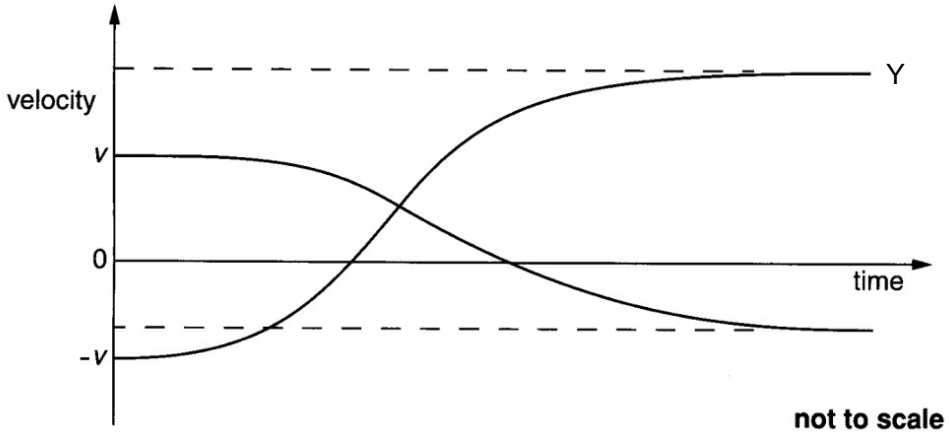


Fig. 2.2

The collision is elastic.

(a) Describe the energy change of the system that takes place during the collision.

……

……

……

……[1]

(b) Explain, using features of the graph, how it may be deduced that X has the larger mass.

……

……

……

……[2]

(c) Determine, in terms of v, the speeds of X and Y after the collision.
Show your workings.

speed of X = ……..

speed of Y = ……..

[4]

[Total: 7]

3 **(a)** Explain what is meant by simple harmonic motion.

……

……

……

…… [2]

(b) Fig. 3.1 shows the variation with time t of the height h above the ground of an object of mass 37 kg that is undergoing vertical simple harmonic motion.

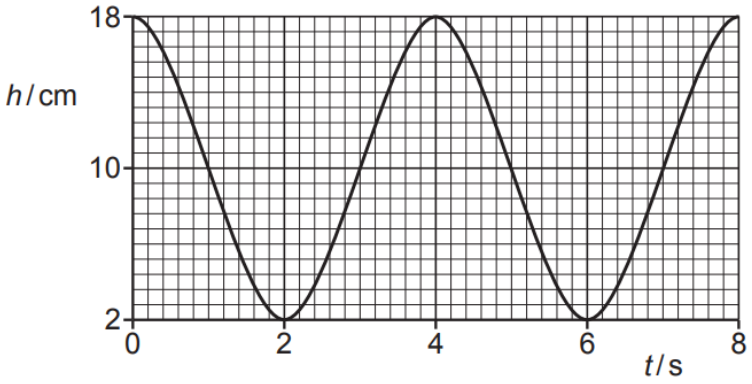


Fig. 3.1

For the oscillations of the object,

(i) determine the amplitude x_0,

x_0 = …… cm [1]

(ii) show that the angular frequency is 1.6 rad s^{-1},

[1]

(iii) determine the total energy E.

E = …… J [2]

(c) On Fig. 3.2, sketch the variation with time t of the kinetic energy E_K of the object.

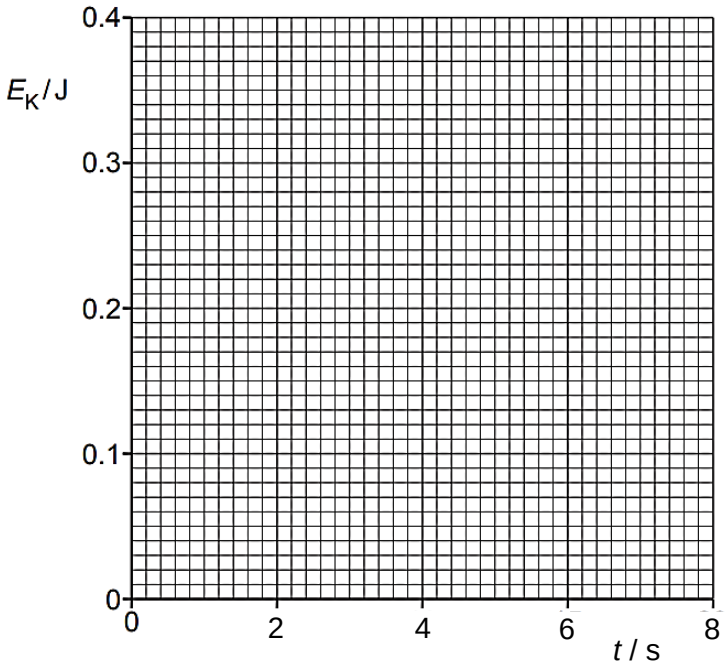


Fig. 3.2

[2]

(d) On Fig. 3.3, sketch the variation with the height h above the ground of the kinetic energy E_K of the object.

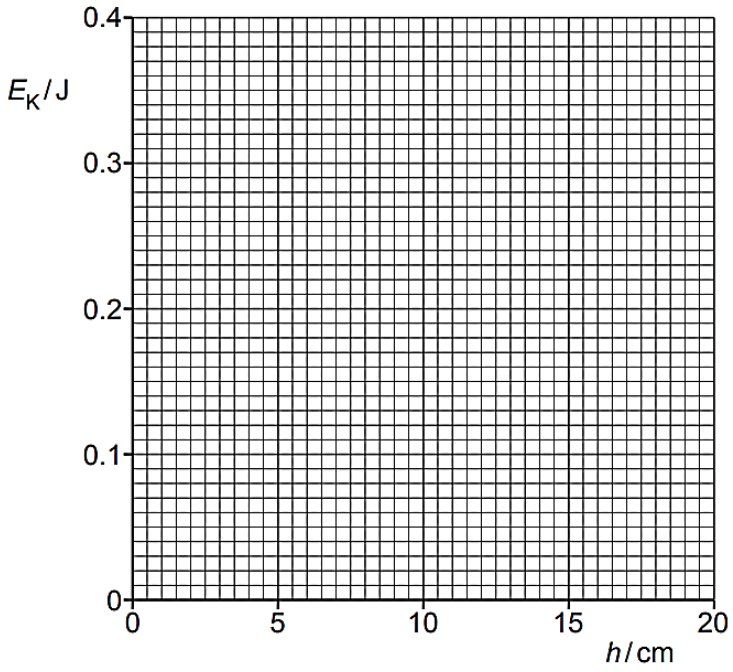


Fig. 3.3

[2]

[Total: 10]

4 **(a)** There is a sinusoidal alternating current in a resistor of resistance 26 Ω. The power P dissipated in the resistor varies with time t is shown in Fig. 4.1.

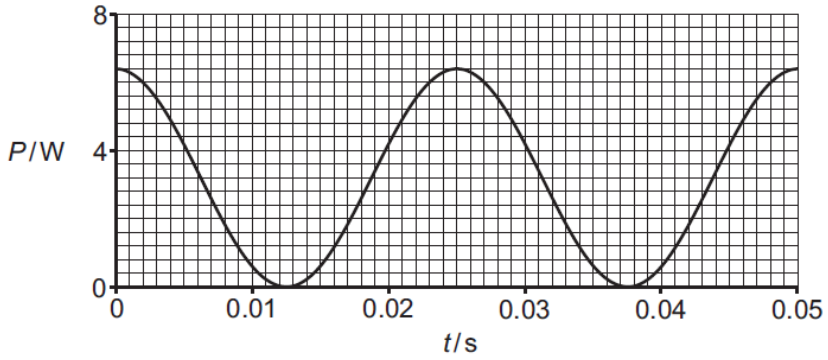


Fig. 4.1

(i) Determine the mean power dissipated in the resistor.

mean power = …….. W [1]

(ii) Calculate the r.m.s. current in the resistor.

r.m.s. current = …….. A [1]

(b) The current in the resistor is now half-wave rectified with the use of an ideal diode.

(i) On Fig. 4.2, sketch the variation of the power P dissipated in the resistor between $t = 0$ and $t = 0.050$ s.

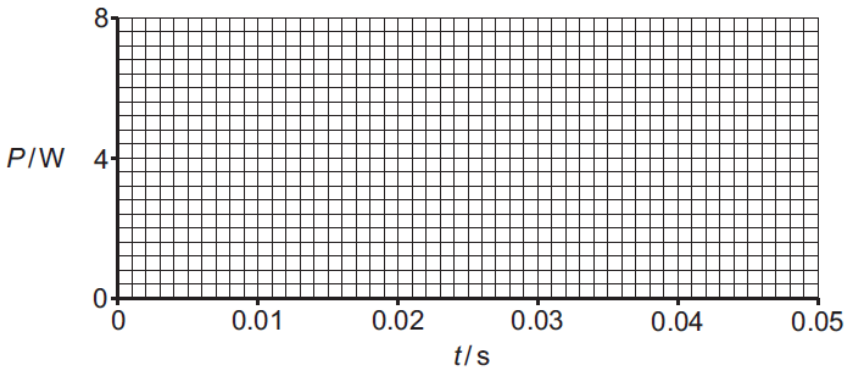


Fig. 4.2

[2]

(ii) Determine the r.m.s current in the resistor. Explain your reasoning.

r.m.s. current = …….. A [2]

[Total: 6]

5 Fig. 5.1 shows a circuit containing a capacitor of capacitance C and a resistor of resistance R.

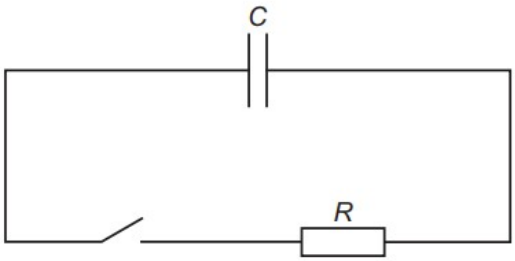


Fig. 5.1

Initially, the switch is open and the potential difference (p.d.) across the capacitor is 12 V.

The switch is closed at time $t = 0$ and the capacitor discharges through the resistor.

Fig. 5.2 shows the variation of the charge Q on the capacitor with the p.d. V_C across the capacitor as the capacitor discharges. Fig. 5.3 shows the variation of the current I in the resistor with the p.d. V_R across the resistor as the capacitor discharges.

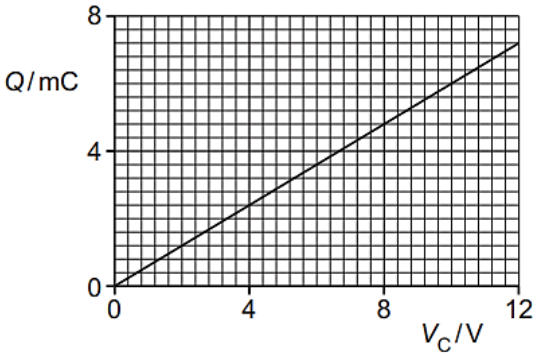


Fig. 5.2

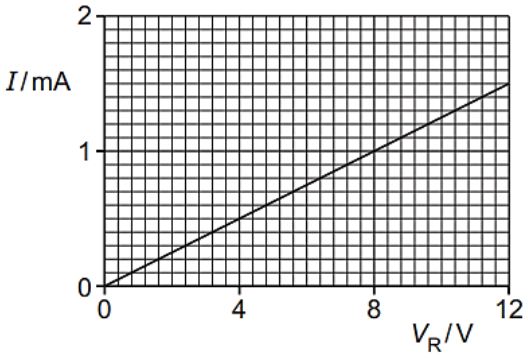


Fig. 5.3

(a) Determine

(i) the capacitance C,

C = …….. F [1]

(ii) the resistance R,

R = …….. Ω [1]

(iii) the time constant τ.

τ = ... s [1]

(b) A quantity is said to vary exponentially when its rate of change is proportional to the quantity itself.

Using Fig. 5.2, Fig. 5.3 and the relationship between V_C and V_R, explain why the variation of Q with t is exponential in nature.

.....

.....

.....

..... [2]

(c) **(i)** The potential difference before the capacitor starts discharging is V_O. Determine, in terms of V_O, the potential difference across the capacitor when the energy stored in the capacitor is half its initial value.

potential difference = ... [2]

(ii) The time taken for the potential difference of the capacitor to be reduced to half its initial value is T. Determine, in terms of T, the time taken for the energy stored in the capacitor to decrease to half its initial value.

time taken = ... [2]

[Total: 9]

6 **(a)** Electrons in a vacuum are accelerated from rest through a potential difference (p.d.) V to form a beam. Each electron has mass m and charge q.

The beam is incident on a graphite crystal that acts as a diffraction grating. After passing through the crystal, the beam reaches a fluorescent screen. An interference pattern is observed on this screen.

(i) Explain what this observation shows about the nature of electrons.

……..

……. [1]

(ii) Determine an expression, in terms of m, q, V, for the momentum p of an electron in the beam.

p = …….. [2]

(iii) The p.d. through which the electrons are accelerated is now increased to a greater value. Describe and explain the effect of this change on the interference pattern observed.

……..

……..

……..

……..

……. [2]

(b) A spherical quantum dot made of a semiconductor can be modelled as an electron confined in a one-dimensional infinite square well of width L as shown in Fig. 6.1. L is taken to be 5.0 nm.

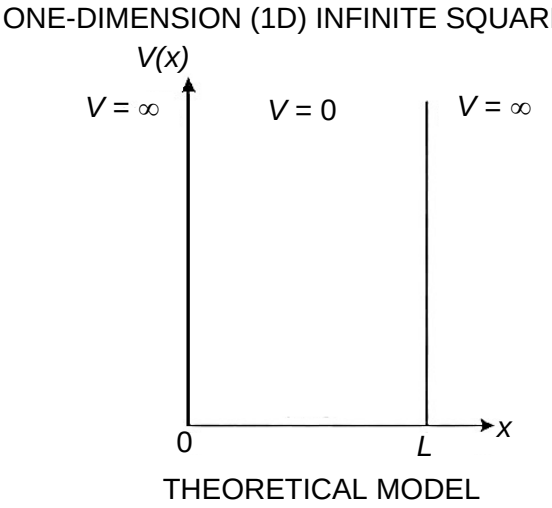


Fig. 6.1

The normalised ground-state wavefunction of the electron inside the well ($0 < x < L$) is:

$$\psi_1(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$$

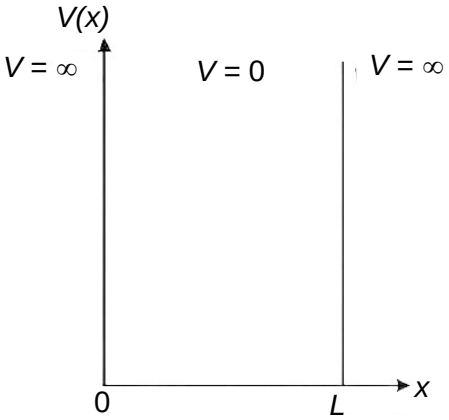
Outside the well, $\psi_1(x) = 0$.

(i) Explain what is meant a normalised wavefunction.

……..

…… [1]

(ii) Sketch the wavefunction in Fig. 6.2.



[1]

Fig. 6.2

(iii) Calculate the ground state energy E_1 of the electron in this quantum dot.

energy = …….. J [1]

(iv) If the electron makes a transition from $n = 2$ to $n = 1$, find the wavelength of the emitted photon.

wavelength = …….. m [2]

[Total: 10]

7 **The Physics of High-Performance Transit: Singapore's R151 MRT Trains**

The Singapore Mass Rapid Transit (MRT) system recently introduced the seventh-generation Alstom Movia R151 rolling stock for the North–South and East–West lines. These trains operate using a 750 V d.c. third-rail electrical supply.

Electrical power from the national grid is supplied to traction substations located along the MRT line. At the Buona Vista (BNV) substation, electrical power is received from the national grid at an alternating potential difference of 22 kV. A step-down transformer reduces this to a r.m.s. potential difference of 750 V a.c., before a rectifier converts the a.c. supply to d.c. for the third rail. The transformer can provide a r.m.s. secondary current of 7200 A.

A single trainset consists of 6 cars in a DT–M1–M2 + M2–M1–DT formation, where 'M' denotes a motored car and 'DT' denotes a trailer car with a driver's desk. To enhance efficiency, these trains are constructed from lightweight aluminium alloy, giving a 6-car set a length of 138.86 m and a width of 3.2 m. The car height is approximately 3.7 m. A fully loaded train has a mass of approximately 300 tonnes where 1 tonne equals to 1.0×10^3 kg.

While designed for a maximum speed of 90 km h^{-1}, the trains typically operate at a service speed of 80 km h^{-1}. To ensure safety, automated systems maintain a constant deceleration of 1.3 m s^{-2} during emergency stops.

Sustainability is a core focus for the Land Transport Authority (LTA). R151 trains utilize high-efficiency traction systems capable of regenerative braking. In this mode, the traction motors act as generators, converting the train's kinetic energy into electrical energy that is fed back into the third-rail for use by other trains on the line. The effectiveness of this recovery depends on the line receptivity—the requirement that other trains must be operating nearby to immediately consume the electricity as it is generated. If no other trains are in the vicinity to use the power, the system is 'non-receptive,' and the excess energy will be dissipated as heat through on-board resistors.

A train in motion experiences total resistance consisting of mechanical friction and air drag. Generally, the resistance W_0 (in newtons) is often modeled by the Davis equation:

$$W_0 = a + bv + cv^2$$

where a, b and c are empirical coefficients and v is the velocity.

For a 300-tonne SMRT train, the specific resistance R (in newton per tonne) in open air is defined as:

$$R = 2.3 + (0.6v^2 \times 10^{-3})$$

where v is the speed in km h^{-1}.

In confined underground tunnels, air resistance is significantly amplified by the "piston effect". As a train moves, it compresses the air column in front of it while creating a low-pressure wake behind it. The magnitude of this effect is governed by the blockage ratio (β), the ratio of the train's frontal area (A_{tr}) to the tunnel's cross sectional area (A_t):

$$\beta = \frac{A_{tr}}{A_t}$$

Research into the high-blockage tunnels has identified a critical transient phenomenon during the deceleration phase:

1. Steady-state motion: At constant velocity, the air resists the train's motion (negative drag), requiring motor power to overcome.
2. Transient deceleration: As the train brakes, the air column trapped behind it continues to move forward due to its own momentum. This air column exerts a positive force on the rear of the train, essentially "pushing" it in the direction of travel.

In the research, the variation of train drag coefficient C_D with time t of scaled model trains with different β values in an experiment is shown in Fig. 7.1.

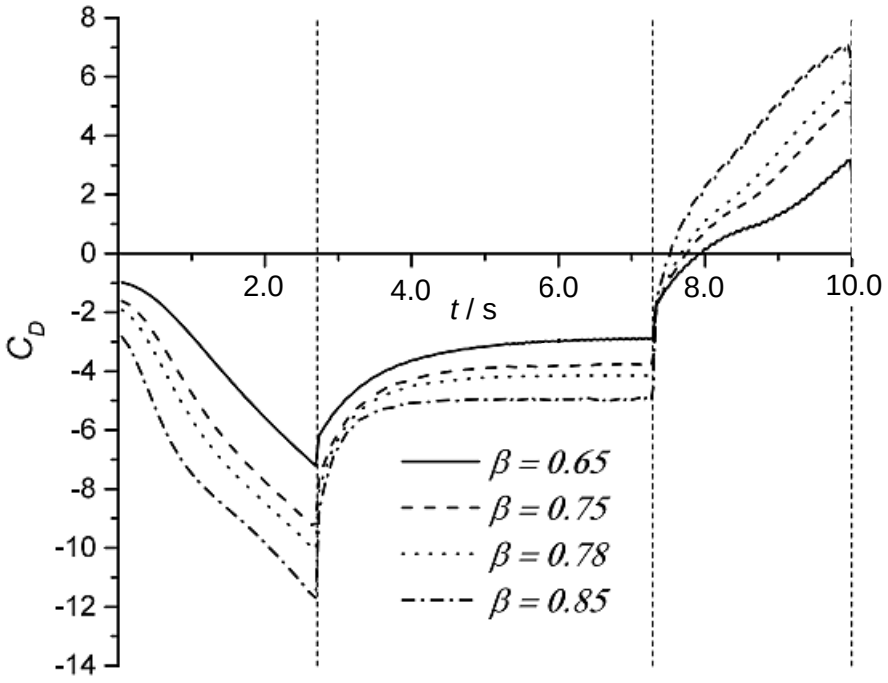


Fig. 7.1

During the experiment, the train first accelerates at 1.0 m s^{-2}, then moves with constant speed and then decelerates in the confinement of a tunnel wall.

The drag coefficient C_D is a dimensionless quantity that characterizes the aerodynamic resistance of an object based on its shape and the fluid environment. In this experiment result, a negative drag coefficient C_D means that the fluid pushes the object backwards while a positive drag coefficient C_D means the fluid pushes the object forward.

(a) Determine the S.I. base units of the coefficient c in the Davis equation.

base units of c = …….. [2]

(b) **(i)** For a SMRT train cruising at service speed in open air, determine the resistance acting on the train.

resistance = …….. N [2]

(ii) In a confined tunnel, the resistance acting on the train cruising at service speed is increased to 42.5 kN due to the "piston effect".

1. Determine the output power of the train to overcome the resistance force.

output power = …….. W [2]

2. Given that the train draws a current of 2100 A, determine the efficiency of the train.

efficiency = …….. % [2]

3. Suggest why the efficiency of the train is not 100%.

……..

……..

……..[1]

(c) For the Buona Vista substation,

(i) calculate the turns ratio (secondary turns to primary turns) for the transformer used in this substation,

turns ratio = …….. [2]

(ii) calculate the peak current drawn by the substation from the 22 kV grid, assuming the transformer operates with 100% efficiency.

peak current = …….. A [2]

(d) **(i)** If the regenerative system captures 30% of kinetic energy of the train, determine the amount of electrical energy produced through the regenerative braking system when a fully loaded train enters a station.

electrical energy produced = …….. J [2]

(ii) Official LTA data states that regenerative braking allows trains to recuperate approximately 30% of the total energy consumed.

Explain why the percentage might decrease during off-peak hours when fewer trains are running on the line.

……..

……..[1]

(e) **(i)** Determine the cross-sectional area of the tunnel if the SMRT wants to design the underground track system to have a blockage ratio of 0.65.

area = …….. m^2 [1]

(ii) In Fig. 7.1, the experimental data shows that the drag coefficient becomes positive at approximately t = 7.5 s.

Explain the physical cause of this “positive force” and its impact on the braking system.

……..

……..

……..

……..[2]

(iii) A researcher claims that increasing the blockage ratio has the most severe impact on the train's braking systems rather than its acceleration systems.

The maximum and minimum values of C_D due to different blockage ratio (β) is shown in Fig. 7.2.

	Max C_D value	Min C_D value	Percentage increase
Peak resistive drag during acceleration	11.8	7.2	64%
Peak forward push during deceleration			

Fig. 7.2

Complete Fig. 7.2 with data from Fig. 7.1 and evaluate the researcher's claim.

……..

……..

……..

……..[2]

[Total: 21]

BLANK PAGE